

MEC 427 Advanced Sensors and Actuators Challenges 1-5

Date & Time

- **Report Date:** 12-01-2025
 - **Time Block:** 12-01 – 12-06
 - **Time Spent This Session:** 12 hours / 0 minutes
 - **Cumulative Time on Project:** 45 hours / 30 minutes
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Project Overview

- **Project Title:** Challenge 4: Cable trolley, and Challenge 3: Water control system.
 - **Project Objective(s):**
 - Design an actuator system that can traverse a steel tension cable.
 - **Current Phase:** Completion
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Task / Activity Details

- **Module / Subsystem:** Challenge 4 Gantry motion control system Wiring and model design refinement
 - **Specific Task or Activity:**
 - **Time Spent on This Task:** 6 hours / minutes
 - **% of Total Time:** 80%
 - **Resources Used:**
 - **Hardware:**
 - **Software:**
 - **Tools:** Grok AI, Fusion 360.
 - **Documentation:** Design Sketches CAD designs
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- **Module / Subsystem:** Challenge 3 Water control system coding and testing
- **Specific Task or Activity:**
- **Time Spent on This Task:** 2 hours / 30 minutes
- **% of Total Time:** 50%
- **Resources Used:**
 - **Hardware:** Adafruit CPX, capacitive touch sensors, servo pinch valve, water tower model.
 - **Software:** VsCode, CircuitPython.
 - **Tools:** Copilot AI

- **Documentation:** AI conversation, code files, test results.

- **Project Status Update:**

- **Task/Module:** Challenge 4 Gantry motion control system design
- **Project Completion Status:** 90% complete
- **Milestones Reached:** ``CAD Model Updated to final design, STL files generated for 3D printing.`
- **Blockers or Issues:** Delays in 3D printing due to printer availability.
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- **Task/Module:** Challenge 3 Water control system coding and testing
- **Project Completion Status:** 100% complete
- **Milestones Reached:** Fine tuned the system to reliably control water flow based on sensor input.
- **Blockers or Issues:** Fine-tuning the sensor calibration to avoid false readings.

Results & Deliverables

- **What Was Achieved:** Completed Challenge 4 Wiring and finished challenge 3.
- **Artifacts Created:** CAD designs for cable trolley system. Code and test results for water control system.
- **Validation / Testing Outcomes:** Water tower actuation tests show reliable operation with current calibration.
- **Changes Made to Design or Approach:** GroK AI was used to assist in design ideation and STL generation with tinkercad.

Narrative & Observations

Use this section for detailed descriptions, reflections, and context. Include technical notes, emotional observations, and rationale for decisions.

- **Narrative Summary:** Challenge 3 was completed with successful unit testing of the software and functionality of the water control system. Challenge 4 design was refined and wiring schematics were developed, but physical assembly is pending due to 3D printing delays. Thinking of my workflow I can see the 'hobbyist' side of me wanting to get my hands on the physical parts. I did not spend enough time considering various aspects of system design. The carriage itself will work hypothetically, things such as housing for electronics and the weight and distribution of components were not fully considered. Also I need to work on my documentation skills to better track changes and decisions made during the design process. Over the coming break I will go over my project assets and develop my overall workflow to be more efficient and effective.